Dynamic Platoon Formation of Multi-Type Autonomous Vehicles for Sustainable Urban Mobility

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Abstract

We study an algorithmic abstraction for future cooperative autonomous mobility in which larger active vehicles can tow smaller passive vehicles along shared road segments. The core problem is spatio-temporal: given fixed vehicle trajectories, entry times, speeds, and active-vehicle towing capacities, determine which passive vehicles should be matched to which active vehicles, and over which route segments, to maximize total covered distance. Unlike standard platooning or one-to-one assignment, the problem allows multi-segment service, where a passive vehicle may be towed by different active vehicles over disjoint portions of its route, while active-vehicle capacity must be respected at every position along the corridor. We formulate this dynamic platoon formation problem and propose two approximation methods: a greedy maximum-weight matching baseline and an Iterative Linear Assignment method that expands active vehicles into virtual capacity slots and repeatedly solves linear sum assignment subproblems. Experiments on synthetic corridor scenarios with up to 400 passive and 80 active vehicles show that both methods achieve substantial towed-distance coverage (25–52% depending on capacity), with the assignment-based method consistently matching or exceeding greedy solution quality while reducing empirical runtime by 2.8–11.5×. We position the model as a corridor-level spatial matching primitive; mechanical coupling, safety, and full road-network routing are left as future extensions.

CCS Concepts

• **Theory of computation** → **Network optimization**; • **Information systems** → *Spatial-temporal systems*; • **Applied computing** → Transportation.

Keywords

autonomous vehicles, vehicle platooning, spatial-temporal matching, linear assignment, sustainable mobility

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1 Introduction

Single-occupancy travel remains a major barrier to sustainable urban mobility. U.S. National Household Travel Survey data show that drive-alone commuting dominates trip demand [1], while the transportation sector contributes about 29% of U.S. greenhouse gas emissions [2]. This inefficiency is pronounced when many vehicles traverse overlapping routes but still consume propulsion energy independently. With rapid advances in autonomous driving [3], future transportation can realize physically coordinated vehicle platoons that go beyond independent driving.

In a near-future setting where vehicles are fully autonomous, new forms of physical coordination between vehicles become feasible: a large autonomous vehicle could mechanically couple with smaller vehicles on a shared road segment, towing them and offsetting a substantial share of their independent propulsion energy [3, 4]. Concrete engineering precedents already exist in patent and public-transport research: a U.S. patent describes autonomous platooning vehicles connected by mechanical linkages controllable by an onboard unit [8], and the Dynamic Autonomous Road Transit (DART) study investigated electronic coupling and decoupling of platooned units along travel corridors [9]. Realizing such coordinated interaction at scale requires *systematic decisions* about which vehicles should couple, when, and for how long along their routes. We therefore treat mechanical coupling, safety, and regulatory design as system assumptions outside the scope of this short paper, and focus on the spatio-temporal matching layer that would be required if such coupling were available.

To address this challenge we introduce a new problem that we call *dynamic platoon formation.* We consider two classes of autonomous vehicles: larger *Active Vehicles (AVs)* capable of physically towing smaller vehicles, and smaller *Passive Vehicles (PVs)* that can be attached to AVs during shared road segments. Unlike drafting-based platooning, which yields modest aerodynamic savings while vehicles remain physically separated, our model assumes mechanical coupling: while a PV is attached to an AV we model the PV as not self-propelled over that segment, with the segment's towing load attributed to the AV. Given a set of AVs and PVs with known, fixed routes (AVs do not deviate from their pre-planned paths), entry times, and speeds, the goal is to **maximize the total PV distance covered by towing**. We use towed distance as a proxy

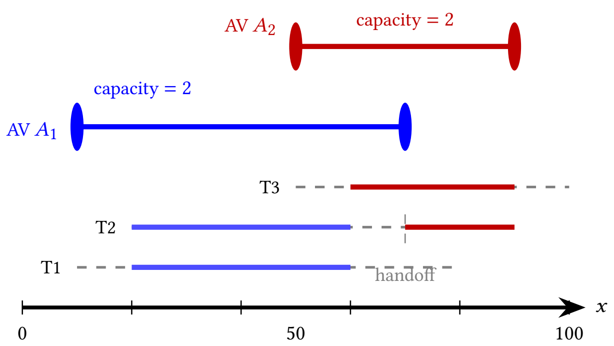


Figure 1: Towing-based cooperation on a 1D corridor (0–100 m). Solid segments are towed travel (PV modeled as not self-propelled); dashed segments are self-driving. AVs A_1, A_2 each have capacity 2; $T2$ is handed off from A_1 to A_2 at x=70, illustrating *multi-segment matching*.

for PV-side propulsion-energy reduction and reserve physics-level load modeling for future work.

Figure 1 illustrates this on a shared corridor with two AVs and three PVs. Two features are central to our formulation: *point-wise capacity*—between x=20 and x=60, A_1 simultaneously tows $T1$ and $T2$, reaching its capacity of 2—and *multi-segment matching*, where $T2$ is served by A_1 and A_2 on disjoint portions of its route with a self-driving gap in between.

Prior work falls into three threads: platooning via aerodynamic coordination among separated vehicles [4–7]; ridesharing and dial-a-ride matching of passengers or trips [10–13]; and combinatorial one-to-one assignment [14, 15]. None captures the combination required for dynamic towing coordination: sustained physical coupling, multi-segment coverage of one PV by multiple AVs, and *point-wise* AV capacity that changes as PVs attach and detach along a route.

We formulate the problem as: given AVs and PVs with fixed routes, entry times and speeds, decide which PVs attach to which AVs over which road segments to maximize the total towed PV distance, subject to spatial overlap requirements, point-wise capacity, and temporal synchronization at coupling points. This yields a state-dependent combinatorial matching problem that differs from a static Generalized Assignment Problem (GAP) [16], because assigning one towing segment can change later temporal feasibility for the same PV. Exact global solvers are therefore impractical at fleet scale, and we propose two approximations. The first, a *Greedy Maximum-Weight Matching*, repeatedly commits to the highest-saving feasible AV–PV segment until no feasible pair remains—fast but myopic. The second, an *Iterative Linear Assignment* (ILA) algorithm, takes a globally coordinated view: in each round it expands every AV into virtual capacity slots, builds a bipartite graph of feasible AV-slot–PV pairings, and solves the resulting *Linear Sum Assignment Problem* (LSAP) optimally via the Jonker–Volgenant method [17]. PV states are then updated to reflect newly covered segments and the round repeats until no further assignment is possible. For tractability we evaluate both algorithms on a one-dimensional corridor abstraction; we discuss extensions to 2D road networks in Section 5.

Contributions. (1) We formalize *dynamic platoon formation* as a spatio-temporal matching problem with multi-segment service and point-wise AV capacity, instantiated on a 1D corridor. **(2)** We develop and compare two approximation algorithms—a greedy maximum-weight baseline and an iterative LSAP-based method with virtual-slot capacity encoding. **(3)** On synthetic corridor scenarios with up to 80 AVs and 400 PVs, ILA matches or exceeds greedy solution quality (25–52% towed-distance coverage depending on capacity) while running 2.8–11.5× faster.

2 Problem Formulation

We formalize *dynamic platoon formation* on a one-dimensional corridor of length L. Restricting attention to a single corridor isolates the spatio-temporal matching structure—segment overlap, point-wise capacity, and temporal synchronization—from routing decisions; extending the formulation to 2D road networks is discussed in Section 5. Throughout, positions are in meters (m), times in seconds (s), and speeds in m/s; towing capacities are nonnegative integers.

2.1 Vehicles and Notation

A problem instance consists of N *Active Vehicles* (AVs) $\mathcal{A}$ and M *Passive Vehicles* (PVs) $\mathcal{P}$ on a corridor of length L (m). Each AV $a_i \in \mathcal{A}$ is described by $(e_i^a, x_i^a, C_i, t_i^a, v_i^a)$, where $e_i^a, x_i^a \in [0, L]$ are entry/exit positions in m with $e_i^a < x_i^a$, $C_i \in \mathbb{Z}_{\geq 0}$ is a *scalar* towing capacity, t_i^a is the entry time in s, and $v_i^a > 0$ is a constant speed in m/s. Each PV $p_j \in \mathcal{P}$ is described analogously by $(e_j^p, x_j^p, t_j^p, v_j^p)$ but has no towing capacity. Two instance-level parameters complete the input: a minimum towing-segment length $L_{\min}$ (m) and a temporal coupling tolerance τ (s). The capacity *ranges* reported in Section 4 are values swept across instances, not per-route vectors for a single AV.

2.2 Feasibility of a Towed Segment

A candidate towing segment is a tuple (a_i, p_j, s, e) with $0 \leq s < e \leq L$, meaning p_j is towed by a_i over $[s, e]$. The maximum realizable overlap of (a_i, p_j) is $[cp_{ij}, dp_{ij}]$ with $cp_{ij} = \max(e_i^a, e_j^p)$ (earliest possible coupling) and $dp_{ij} = \min(x_i^a, x_j^p)$ (latest possible decoupling); its length is $d_{ij} = \max(0, dp_{ij} - cp_{ij})$, and any selected segment must satisfy $cp_{ij} \leq s < e \leq dp_{ij}$. The segment is *feasible* if (i) $[s, e] \subseteq [cp_{ij}, dp_{ij}]$, (ii) $e - s \geq L_{\min}$ (minimum length, justifying coupling overhead), and (iii) the two vehicles reach s within tolerance τ:

$$\left| T_i^a(s) - T_j^p(s) \right| \leq \tau, \tag{1}$$

where $T_i^a(x) = t_i^a + (x - e_i^a)/v_i^a$ is AV a_i's arrival time at position x. The PV arrival time $T_j^p(s)$ is *state-dependent*: for an initially uncovered PV it is the self-driving time $t_j^p + (s - e_j^p)/v_j^p$, but after p_j has been towed on an earlier segment, our algorithms (Section 3) update its route-state time before evaluating later segments. We use the towed distance $e - s$ directly as the saving proxy on a feasible segment, reserving energy-level modeling that accounts for AV-side towing load for future work.

2.3 Multi-Segment Matching and Optimization

Let $\mathcal{X}$ denote a set of feasible segments selected by the algorithm. We allow a single PV to be served by multiple AVs along disjoint portions of its route—*multi-segment matching*—but the same PV cannot be towed by two AVs at the same position. The optimization problem maximizes total towed distance

$$\max_{\mathcal{X}} \sum_{(a_i,p_j,s,e)\in\mathcal{X}} (e-s) \tag{2}$$

subject to the *point-wise capacity* requirement

$$\left|\{(a_i,p_j,s,e)\in\mathcal{X} : x\in[s,e)\}\right| \le C_i \quad \forall a_i\in\mathcal{A},\ \forall x\in[e_i^a,x_i^a], \tag{3}$$

together with two further conditions: (ii) segments assigned to the same PV are pairwise disjoint, and (iii) each $(a_i,p_j,s,e)\in\mathcal{X}$ is feasible in the sense of Section 2.2. The objective (2) is the total towed distance across all selected segments, and constraint (3) enforces that at every position x along an AV's route the number of PVs currently attached to it is bounded by C_i.

Problem structure. The problem has a combinatorial assignment structure with *state-dependent* feasibility: selecting one towing segment changes a PV's residual route timing and can affect the temporal feasibility of later segments, distinguishing it from a static GAP-style assignment [16]. We therefore do not pursue an exact global solver in this short paper and instead study scalable approximation algorithms in Section 3.

Critical-points reduction. Although constraint (3) ranges over a continuum of positions x, the count of attached PVs is a step function that changes only at the endpoints of selected segments. It therefore suffices to enforce capacity at the finite set of *critical points* $\mathcal{C}_i=\{e_i^a,x_i^a\}\cup\{s,e:(a_i,p_j,s,e)\in\mathcal{X}\}$ for each AV a_i. Both algorithms in Section 3 exploit this reduction so that each feasibility check costs $O(|\mathcal{C}_i|)$ rather than continuous verification.

2.4 Modeling Assumptions

We adopt the following modeling choices, each open to relaxation in future work. Routes, entry times, and speeds are known and deterministic, and AVs do not re-route to attract PVs; while attached, a PV follows the AV's speed. Coupling and decoupling are modeled as instantaneous and cost-free at the algorithmic layer, with the physical overhead partly absorbed by the minimum-segment requirement $L_{\min}$. Because matching does not change the total vehicle population, we do not model induced congestion in this short paper, and mechanical, safety, and regulatory aspects of physical coupling remain system-level assumptions outside our scope (cf. Sec. 1).

3 Proposed Algorithms

Both algorithms maintain a PV *route-state* $(\ell_j,\hat{t}_j)$ where ℓ_j is the start of the unprocessed suffix of p_j's route (initially e_j^p) and $\hat{t}_j$ its arrival time at ℓ_j (initially t_j^p). When p_j is committed to a segment $[s,e]$ on AV a_i, the state advances by $\ell_j\leftarrow e$ and $\hat{t}_j\leftarrow\hat{t}_j+(s-\ell_j^{\text{old}})/v_j^p+(e-s)/v_i^a$, reflecting self-driving to s followed by towing at the AV's speed to e. The state-dependent arrival time T_j^p in Eq. (1) reads from this $\hat{t}_j$.

3.1 Greedy Maximum-Weight Matching

The greedy baseline repeatedly builds the current feasible segment set using the critical-point capacity check of Sec. 2.3. It commits the segment with the largest $e-s$, updates the route-state, and stops when no feasible segment remains. Each iteration costs $O(NM\log NM)$ for candidate sorting, giving $O((NM)^2\log NM)$ overall; the baseline is fast but myopic, as a high-saving commit can block a co-optimal alternative.

3.2 Iterative Linear Assignment (ILA)

ILA replaces the myopic commit with a sequence of *global* matching rounds. In each round, every AV a_i is replicated into C_i identical *virtual slots*, encoding per-AV multiplicity in the LSAP while leaving the position-dependent capacity check to the commit step. With $K=\sum_i C_i$ slot nodes on the left and M PV nodes on the right, we form a cost matrix $W\in\mathbb{R}^{K\times M}$ whose entry for slot k of AV a_i and PV p_j is $-\bar{d}_{ij}$ for a feasible pair and $+\infty$ otherwise, where $\bar{d}_{ij}=\max(0,\ dp_{ij}-\max(e_i^a,\ell_j))$ is the residual overlap that uses ℓ_j in place of e_j^p. Minimum-cost matching on W therefore maximizes total towed distance for the current round.

Each round (Algorithm 1) solves this Linear Sum Assignment Problem (LSAP) optimally with the Jonker–Volgenant algorithm [14, 17]. Returned pairs are then sequentially confirmed against point-wise capacity at the critical points; pairs that fail (because multiple slots of the same AV matched a non-disjoint set of segments in one round) are discarded and remain eligible in later rounds. Confirmed pairs advance the corresponding PV route-states, so the next round operates on the residual problem. Rounds repeat until no feasible match remains, yielding multi-segment service whenever the same PV is matched to different AVs across rounds. Jonker–Volgenant guarantees per-round optimality on the constructed bipartite subproblem; global optimality across rounds is not guaranteed, but our experiments (Section 4) show ILA matches or exceeds greedy on every tested scenario. Each round runs in $O(KM+\max(K,M)^3)$, and at most $O(NM)$ rounds are required since each commit strictly advances some ℓ_j, giving overall complexity $O(NM\cdot\max(K,M)^3)$.

Algorithm 1 (ILA).

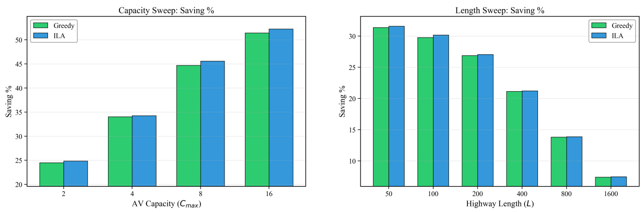
Require: AVs $\mathcal{A}$, PVs $\mathcal{P}$, $L_{\min}$, τ
Ensure: $\mathcal{X}$

```
(ℓ_j, t̂_j) ← (e_j^p, t_j^p) ∀j;  X ← ∅
repeat
    Build W on K = Σ_i C_i virtual slots;  M ← LSAP(W)
    for (slot: a_i, p_j) ∈ M with W_{slot,j} < +∞, in order do
        s ← max(e_i^a, ℓ_j),  e ← dp_ij
        if (a_i, p_j, s, e) feasible w.r.t. prior commits this round then
            X ← X ∪ {(a_i, p_j, s, e)};  update ℓ_j, t̂_j
        end if
    end for
until no commit in the last round
return X
```

4 Experimental Evaluation

4.1 Setup

Both algorithms are implemented in Python 3.11 with NumPy 1.26 and SciPy 1.11 [17]; ILA's per-round LSAP uses SciPy's compiled Jonker–Volgenant routine. Experiments run single-threaded on an Apple M4 Pro (16 GB unified memory, macOS 15). For synthetic corridor instances, AV and PV entry/exit positions are sampled



(a) Capacity sweep (L=100 m, N=50, M=200).

(b) Length sweep (N=80, M=400, $C_{\max}$=3).

Figure 2: Saving % under (a) varying AV capacity upper bound $C_{\max}$ and (b) varying corridor length L. ILA matches or exceeds Greedy in every tested configuration. Bars are 4-seed means.

uniformly from $[0, L]$, entry times uniformly from $[0, W]$ over a scenario time window W, and speeds uniformly within ±20% of a unit reference. Default parameters are L=100 m, N=50, M=200, $L_{\min}$=10 m, τ=5 s, W=100 s, with C_i drawn uniformly from $\{1, 2, 3\}$. We report means over four seeds (42–45) and run two sweeps. The *capacity sweep* tests $C_{\max} = 2, 4, 8, 16$ at default N, M, L, sampling C_i uniformly from $\{1, \ldots, C_{\max}\}$. The *length sweep* tests six lengths from 50 to 1600 m at N=80, M=400, $C_{\max}$=3, with W=L s so that vehicle density per unit time stays comparable. The headline metric is *saving %*, the total towed distance divided by the no-towing baseline $\sum_j (x_j^p - e_j^p)$; we also report wall-clock runtime.

4.2 Results

Figure 2 reports both sweeps. As $C_{\max}$ grows from 2 to 16, saving % rises from 24.5% to 51.4% for Greedy and from 24.8% to 52.3% for ILA, an ILA advantage of 0.2–0.9 percentage points across the four capacity settings; as L grows from 50 m to 1600 m, saving % falls from 31.4% to 7.4% (Greedy) and 31.6% to 7.5% (ILA), reflecting the diminishing pairwise overlap on longer corridors. ILA matches or exceeds Greedy in every tested configuration. The two algorithms are close in solution quality on a 1D corridor because the linear route structure restricts the combinatorial diversity of feasible matchings; in 2D road networks with richer overlap patterns we expect Greedy's myopic commits to block more globally superior assignments, widening ILA's margin. Wall-clock runtime in our implementation is 2.8–11.5× lower for ILA on the capacity sweep and 5.5–26.9× lower on the length sweep, attributable to SciPy's compiled LSAP solver and to Greedy's repeated $O(NM \log NM)$ candidate sorting; we report this as an empirical implementation observation rather than a theoretical claim, since worst-case complexity favors Greedy (Section 3). Two limitations bound our interpretation: (i) towed distance is a proxy for PV propulsion-energy reduction and ignores the AV-side towing load, and (ii) the offline, full-information setting assumes deterministic trajectories. Section 5 discusses the 2D extension and a stochastic online formulation as the natural next steps.

5 Conclusion and Future Work

We formalized *dynamic platoon formation* as a spatio-temporal multi-segment matching problem with point-wise AV capacity and showed that an iterative LSAP-based solver matches or exceeds a greedy baseline on every tested 1D corridor configuration. The natural next step is a 2D road-network extension: each PV then has a set of feasible paths rather than a fixed one, lifting the formulation from segment overlap on a single corridor to overlap on a graph with shared edges, which couples our matching layer with route choice. Beyond geometry, two algorithmic openings are interesting at SIGSPATIAL: a stochastic online variant where entry times and routes are revealed over time, and an energy model that internalizes the AV-side towing load so that the objective trades PV savings against AV overhead rather than treating distance as a free proxy.

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